

FOODwell

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ABSTRACT

Hunger is an often neglected issue, with millions of people worldwide affected by it in 2021 according to the United Nations. This issue also affects minority populations in Taiwan. One solution to this problem is the implementation of suspended meals, which are free meals provided to those in need. Our proposed solution, FOODwell, aims to allow recipients to dwell on better food provision and obtain wellness for their food needs. It is a system that simplifies the process of reserving and collecting suspended meals for recipients by enabling them to interact with the kiosk at the convenience store and cashier at the restaurant, without compromising confidentiality or requiring electronic devices. Overall, FOODwell eliminates obstacles for recipients when reserving and collecting suspended meals.

CCS CONCEPTS

• Human-centered computing → User centered design.

KEYWORDS

interaction design, user research, suspended meals, zero hunger, kiosks

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1 INTRODUCTION

Food is essential to life, but for some people in our society, it is still a luxury they can't afford. The United Nations sustainable development goals of "zero hunger" and "reduced inequality" both

reveal the urgent need for people to receive enough food and improve their nutrition inequality. Many countries have sought to solve this problem. One of the approaches is to provide suspended meals—meals that were paid in advance or offered by restaurant owners in order to serve people who are in need or could not afford them. According to the UN, over 3 billion people lacked access to a healthy diet in 2020, and 7 to 8 million people were affected by hunger globally in 2021 [2]. Similarly, in Taiwan, there is a minority population that suffers from food insecurity and would benefit from suspended meals, including but not limited to low-income families, the homeless, the disabled, and the elderly living alone. According to Taiwan's Ministry of Health and Welfare [6], more than 30,000 people are from low-income families and are part of this minority population. However, since the concept of suspended meals was introduced in Taiwan in 2013, suspended meal platforms such as eFOOD Sharing Map¹, Good Day², and Vegemap³ have emerged but suspended food has yet to be fully utilized as intended. To investigate what caused the inadequate leverage of suspended food, we conducted an interview study. Our research uncovered three major issues that contributed to a lack of leverage: 1) Meals are not always secured and saved for suspended meal recipients (SMRs) even after they have waited for a long time; 2) the use of existing platforms requires SMRs devices to browse suspended-meal information, which is prohibitively expensive to them; and 3) SMRs are hesitant to take suspended food due to serious concerns about being recognized as a minority when receiving food by the surrounding people in the restaurant.

We designed the FOODwell service to aid SMRs in receiving their intended suspended meals. It addresses the aforementioned three concerns. FOODwell, in particular, utilizes existing kiosks at every convenience store to enable SMRs to reserve suspended meals without the need for a device. This reservation secures SMRs suspended meals and allows them to know when and where to pick up the meals without having to wait at a restaurant, thereby reducing the risk of being recognized in the restaurant. Meal collection requires only a FOODwell-generated meal receipt, which resembles standard receipts kept by non-SMRs. This strategy ensures

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¹eFOOD Sharing Map, <https://efoodmap.org/>.

²Good Day, <https://goodday.ntpc.gov.tw/pwntpc/>.

³Vegemap, <https://vegemap.merit-times.com/>.

that the food-receiving qualifications of SMRs remain confidential and indistinguishable from those of regular consumers. Once these concerns are resolved, we believe that SMRs will utilize FOODwell more frequently, allowing the welfare to be utilized more efficiently.

2 RELATED WORK

In order to investigate the reasons why suspended meal platforms in Taiwan are not being fully utilized as intended, we begin with identifying three suspended meal platforms that are striving to address the issue of hunger and closely examined their website functions.

2.1 eFOOD Sharing Map

The most crucial function of the *eFOOD Sharing Map* [1] is to allow SMRs to search for the nearest location to obtain food or supplies through an operation similar to Google Maps. By entering SMRs' locations, selecting projects, and whether to receive supplies or suspended meals, SMRs can find restaurants that meet the requirements. After selecting a suitable restaurant, related information and thresholds such as meal coupons only will be listed in the information column. However, since the number of remaining meals is not updated in real-time and there is no detailed explanation of how to reach the thresholds, SMRs may need further help. Moreover, the location of the restaurant is displayed on the map on the right side of the webpage, it can be redirected to Google Maps in order to help SMRs to reach the restaurant easier. If any restaurant wants to register for participation, they can do so through a Google form, or register for related projects.

2.2 Good Day

The *Good Day* platform [3] was established in 2019 with the aim of matching donors and recipients. The platform is donor-oriented, and by accessing the donation page within the platform, the current charity programs led by New Taipei City will be displayed, such as cultivating rising New Taipei City young sports players and suspended meals donation. A comprehensive donation page will be brought up where donors can make donations after clicking on any program. However, the website does not have any links for restaurants to register their participation, and SMRs cannot directly search for suspended meal restaurants through the *Good Day* platform. SMRs must click on another link within this platform, which will be redirected to the *eFOOD Sharing Map* where they search for restaurants that collaborate with the *Good Day* platform.

2.3 Vegemap

Vegemap [7] is a public welfare project initiated by the Buddha's Light International Association R.O.C. in 2020. It not only helps vegetarian restaurants to survive during Covid-19 but also helps people who are in need of food. *Vegemap* provides different access for SMRs, restaurants, and donors. SMRs can filter restaurants based on the address and number of remaining meals, select from two pick-up times with broad time ranges (the very day or the next day between 12:00 AM to 13:30 PM), and register their personal information. SMRs then go to the restaurants and inform the counter to get the meals by the designated time. Additionally, restaurants can apply to join the project by clicking on the link on the website.

After joining, they need to enter the numbers of daily-provided suspended meals and update the remaining meal quantity in real time. Moreover, donors can use the information on the website to donate to a specific restaurant or visit there in person to show support. Although the *Vegemap* project cannot be searched independently on *eFOOD Sharing Map*, restaurants that participate in the project can be seen when searching.

3 USER RESEARCH

3.1 Study Procedure

We did a semi-structured interview study with both restaurant owners and SMRs to investigate why suspended meals did not have enough leverage. We interviewed twenty people (10 males and 10 females) with an age range of 20 to 58 years old. Eleven of them are caring about the hunger crisis, and six of them belonged to the SMR population, denoted as P14 to P20. Among these six actual SMRs, four of them come from low-income families, and the remaining two are unemployed and receiving welfare benefits due to the impact of the COVID-19 pandemic.

We intended to study these individuals' perspectives on receiving suspended meals. The remaining three participants were owners of restaurants that provided suspended meals, labeled as P11 to P13. We inquired as to their knowledge and opinions regarding the preparation and service of suspended meals. We asked both groups of participants about the suspended food systems and platforms that were already in place, including their pros and cons and any problems they had run into. We used affinity diagramming [5] to analyze the interview data, and the primary themes that emerged from the analysis are discussed in the section below.

3.2 Findings

3.2.1 Fear of being identified as a member of a minority group when picking up meals. Our informants told us that the major concern that made many SMRs reluctant to wait in the restaurant and pick up meals at the cashier was being identified as a member of a minority group. This would make them feel that they were being viewed differently by other customers in the restaurant, as P5 said, *"I hope that others will not know my identity and that I will have less psychological burden"*. P6 also noted, *"I don't want to be perceived as someone who simply requires assistance or cannot afford a meal."* P13 also expressed concern that waiting in line would cause them to be perceived differently, *"There can be a queue that stretches over two blocks, and the elderly tend to arrive early to secure a good spot in line. The locals have become accustomed to the queues, but they view the waiting customers with suspicion and therefore choose alternative routes to avoid them."* P17 shared her experiences that she would deliberately avoid the same lunchtime and restaurants as her classmates in the school cafeteria in order to keep her weaker financial situation concealed from them, *"When getting suspended meals at the school cafeteria, I would look for reasons to pick up meals at a separate restaurant from my peers and check if anyone I recognized was nearby, out of fear that my financial circumstances would make me unappealing to them."*

In response to this issue, one participant suggested utilizing food delivery services so that they do not have to pick up the meals at the restaurant, thereby reducing their exposure in the restaurant, as

P5 stated, *“Food delivery is a good solution because I can wait for the meals at home, and I don’t feel embarrassed to be helped by others.”*. However, this option was perceived as expensive and unsuitable for participants who were restaurant owners. Specifically, they mentioned that the current operating cost is already a burden to them, and thus they were reluctant to add the delivery service by themselves, as P11 noted, *“There is no way to do delivery because we don’t have enough resources to do this, and it has been very difficult to maintain the daily meals service.”*

3.2.2 Being worried about not getting meals even after a long wait. It was reported that many SMRs in Taiwan have to wait in long lines during mealtimes because of the current method of distributing suspended meals. Unfortunately, despite having waited a long time for their meals, there was no assurance they would be served because supplies were low, one SMR stated, *“Collecting the free meal often takes me up to an hour, and due to the limited quantity, many people may find themselves unable to receive their meals.”* (P20). P18 also shared his sad experience, *“When I was a child, my mother asked me to collect surplus meals for the family. After a long wait, the food had already been given out. I felt guilty and kept blaming myself.”*. This may have resulted in a loss of opportunity; if they had waited elsewhere or used the time to do anything else, they would not have become hungry during and after the wait. The current system makes it difficult for our participants to get suspended meals, as P20 stated *“I once tried to reserve suspended meals online from a platform, but because their system wasn’t very well-developed, there was none left for me to collect when I went to pick up the food in person.”*. If they were given assurance about the meals, or at least timely information on where and when they can or can’t get them, they could use the wait time more efficiently and wouldn’t have to compete for a good spot in the line.

3.2.3 Lack of verification can lead to abusive meal reservations. Last but not least, some restaurant owners were concerned about non-SMRs abusing the system, and some SMRs were unable to endorse abusive behavior, as P16 mentioned, *“I have seen many people are being opportunistic and taking multiple portions of meals at once.”*. P20 also complained, *“Currently, there is a considerable number of individuals who do not meet the eligibility criteria but are picking up the meals, and some even do so repeatedly. However, the restaurant owners do not intervene due to their age.”*

That is, while it is deemed critical to avoid suspended meals being picked up by non-SMRs, verifying the meal recipients’ qualification or eligibility for obtaining the meal at the cashier would unavoidably expose the meal transaction to nearby customers, rendering the SMR’s reasons for taking the meal non-confidential. To protect the SMRs, one restaurant owner stated, *“Usually, we will not ask him to show his ID as a suspended meal recipient, we will treat them as a general guest.”* (P12). However, it is critical to avoid the system being abused by ensuring that meals intended for SMRs are not taken by non-SMRs, as this would result in increased costs for the restaurant, as P11 stated, *“We can’t waste the kindness of the public and let you pick up meals at will. If there is no verification, some people will abuse it.”*

4 DESIGN REQUIREMENTS

Based on the aforementioned key findings, we generate three design requirements:

- **Ensure that SMRs are not distinguishable from others:** To prevent SMRs from being identified as different from others, which hinders their willingness to use the suspended meals system, it is essential to make them blend in with others. This can be achieved by avoiding long lines that draw attention and making the process of obtaining suspended meals indistinguishable from others, thus reducing the likelihood of others noticing and paying attention to SMRs while they pick up the suspended meals.
- **Inform SMRs of availability of suspended meals and allow a reservation to avoid queuing and uncertainty:** To ensure efficient use of suspended meals and minimize wait times for SMRs, the system should provide information on available meals, allow for reservations, and schedule pickup times for SMRs. This will help distribute SMRs throughout the day and reduce uncertainty for SMRs about obtaining a meal. With this approach, SMRs who come to the restaurant will not leave empty-handed.
- **Check qualification and eligibility for reserving suspended meals:** To ensure suspended meals are reserved by the intended recipients, the system should verify the qualifications and eligibility of those reserving the meals to prevent non-SMRs from reserving them.

5 DESIGN PROCESS

Our group conducted ideation sessions and created various design concepts and variations, keeping in mind the design requirements mentioned earlier. Our goal was to develop an integrated platform that connects SMRs with restaurants willing to provide suspended meals, balancing the needs and addressing the concerns of both parties. We consolidated group members’ sketches of ideas and proposed design ideas and created low-fidelity prototypes for them. Our initial ideas focused on the flow and key features of the system, such as identity verification, meal reservations, and cashier transactions. We received feedback on these prototypes from various groups and potential users and made adjustments based on their input. One of the major changes in our design process was the realization that some SMRs do not have regular access to computers or smartphones, as one of the potential users indicated that *“Some people in need of food assistance, such as homeless people, may not have smartphones or unlimited internet access. Therefore, developing systems on mobile phones is difficult to reach them. Additionally, they may have lower proficiency in using smartphones.”* This prompted us to look for alternative technological solutions that are more widely available and convenient for them to use on a daily basis. As a result, we redesigned the SMRs’ interaction with the platform to incorporate these solutions.

In response to the discovery that some SMRs do not have regular access to computers or smartphones, we decided to utilize existing interactive kiosks that are commonly used by Taiwanese people in



Figure 1: Interactive kiosks that are commonly found in convenience stores in Taiwan, such as ibon, FamiPort, Life-ET, and OK.go

their daily lives, such as ibon⁴, FamiPort⁵, Life-ET⁶, and OK.go⁷, found in convenience stores (see Figure 1). These kiosks are commonly found in most convenience stores in Taiwan and are used by residents to perform various tasks such as cloud printing, paying fees, and picking up tickets. They are equipped with a scanner and chip reader that can read identity information, such as the National Health Insurance Card (NHI Card) [4], which residents in Taiwan typically possess. As a result, these kiosks have been used to verify eligibility and identity for reserving a variety of items and services by inserting the NHI Card⁸ into the kiosk (marked in Figure 1). For example, during the early pandemic period in Taiwan, the government used them to verify residents' eligibility for reserving masks when they were in short supply [9]. These kiosks can also print receipts or tickets for people to bring to cashiers or designated shops to pick up the reserved items. As convenience stores are widely available in Taiwan and are accessible 24/7, they are highly suitable portals for SMRs who have limited technology access to use our system. By leveraging these kiosks, our new design performs eligibility verification and meal reservations assuming that SMRs' eligibility for receiving suspended meals can be verified through the health system in Taiwan. Distributing the verification process to these kiosks, not only saves time for the restaurant's cashier but also prevents SMRs from being checked by cashiers which reduce attention from nearby people.

As a result of this change, we redesigned our system to display the meal reservation interface and the verification process on the kiosks. The restaurant side focuses on viewing and confirming reservations, updating the status of reservations, checking the availability of remaining meals, and scanning the QR code generated by the kiosk for SMRs to pick up the meal, keeping the interaction between SMRs and the restaurant simple - transaction based on the

receipt. We named the system "FOODwell" to stress that recipients can obtain wellness for their food needs through this system, and enjoy better food provision.

6 THE SYSTEM: FOODWELL

FOODwell is a digital service that utilizes existing kiosks in convenience stores as a means for SMRs to reserve suspended meals. The service allows SMRs to pick up their reserved meals without revealing their status as an SMR, by simply showing cashiers at the restaurant counter a receipt generated from a kiosk after verification, which looks similar to regular receipts. The process of using FOODwell service is illustrated in Figure 2. At a convenience store, SMRs can reserve suspended meals by using the kiosk. On FOODwell, they can select preferred meals and pick-up times offered by the restaurant. FOODwell distributes the pick-up times of SMRs evenly, to avoid having many SMRs pick up food at the same time, reducing the likelihood of nearby people noticing them. To complete the meal selection on the kiosk, SMRs are required to insert their National Health Insurance Card (NHI Card) into the kiosk for verification of qualifications and eligibility. After verification, the SMR receives a receipt with a QR code and assigned pick-up time on it. The receipt will resemble the look of a regular receipt of the restaurant, where SMRs pick up the meals. Once the reservation is made, the restaurant receives a notification and confirms it. The system automatically updates the number of remaining meals, which is in real-time synchronized with kiosks. When the SMRs arrive at the restaurant, they present the receipt to the cashier without fear of being recognized, as their transaction appears similar to that of regular customers. FOODwell minimizes SMRs' fear of disclosing their identity and reduces the restaurants' effort to verify their qualifications, as this has already been done at the kiosk. The following sections provide an in-depth look at the FOODwell interfaces in steps.

6.1 Reserving Suspended Meals from Interactive Kiosks at Convenience Stores

We design the FOODwell to be available on interactive kiosks located in convenience stores in Taiwan. By integrating the service into these kiosks, we make the reserving of suspended meals as convenient as possible. As shown in Figure 3 (a, b), we take ibon's interface as an example for the following description. The system is easy to use and requires no prior knowledge or tutorials. The recipients can choose from a list of nearby restaurants and select the meals they want at specific time slots. The remaining number of suspended meals available is also displayed to ensure that SMRs are able to reserve a meal. After completing meal selection, the system prompts the SMRs to insert their National Health Insurance Card (NHI Card) into the machine for qualification check. The NHI Card is equipped with an IC chip that holds information about a person's health and financial situation. During the COVID-19 pandemic in Taiwan, a real-name rationing program for masks and rapid COVID-19 test kits used the NHI Card to verify individuals' qualifications. By inserting the NHI Card into a kiosk, the system can check if the user who operates the system is qualified as an SMR to receive suspended meals, according to government data. Once eligibility is confirmed, a receipt is printed with a reservation

⁴ibon, <https://www.ibon.com.tw/>.

⁵FamiPort, <https://www.famipor.com.tw/>.

⁶Life-ET, https://www.hilife.com.tw/serviceInfo_lifeET01.aspx.

⁷OK.go, https://www.okmart.com.tw/okGo_service.

⁸National Health Insurance Card (NHI Card), <https://eng.nhi.gov.tw/en/cp-22-03586-16-2.html>.

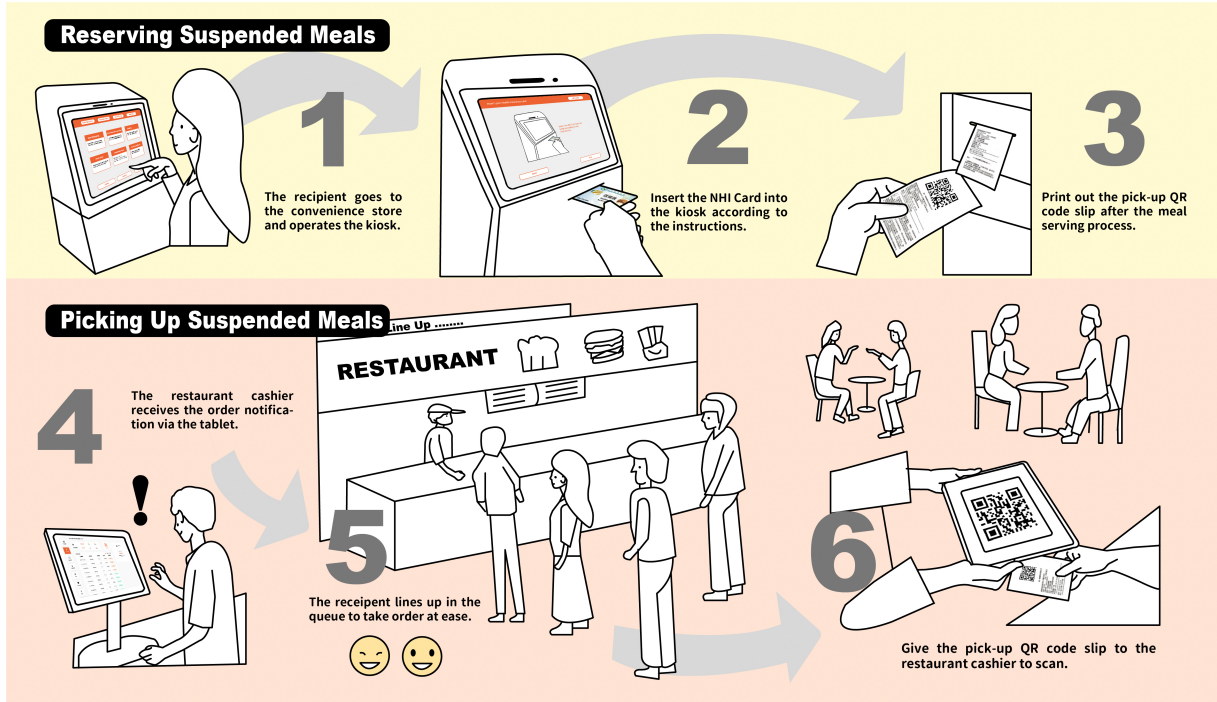


Figure 2: FOODwell workflow.

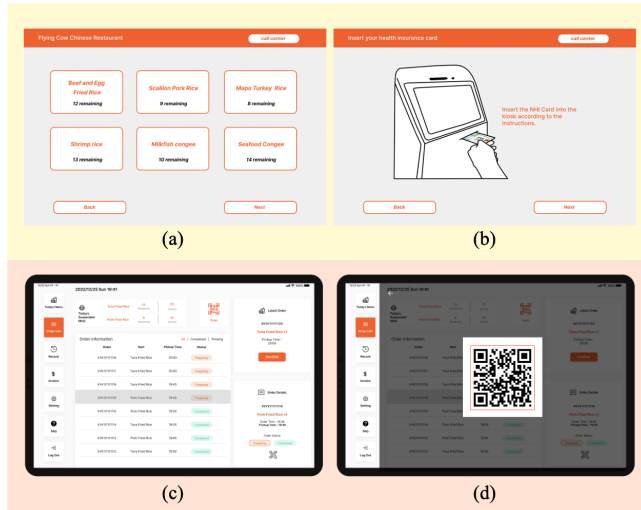


Figure 3: FOODwell interfaces on the interactive kiosk (a, b) and restaurant tablet (c, d).

number, meal item, restaurant address, pick-up time, and a QR code for scanning to complete the transaction.

6.2 Picking up Suspended Meals at Restaurants

As more people are using food ordering platforms, we have noticed that restaurants are becoming more accustomed to using tablets for managing their meal orders. In response to this, our interface

design has been tailored to the use of tablets and is divided into three columns, with one navigation sidebar on the left, a main function area in the center, and another sidebar on the right. The left sidebar features a tab labeled "Order Info", which is specifically focused on meal ordering and pickups and will be the main topic of discussion.

The "Order Info" is the landing page of the FOODwell system on the restaurant side to help keep track of meal orders. Figure 3 (c, d) depicts two steps of the Order Info interface, from receiving meal orders to distributing meals when the SMRs come to pick them up. As soon as a new order is created from the kiosk, a "latest order" notification will appear to alert the restaurant cashier. Then, the new order will be added to the order information list that is sorted by the estimated pickup time. The cashier will change the status of the meal from "prepared" to "completed" once the meal is ready. When the SMRs arrive at the restaurant counter, the cashier can confirm the meal by either the order number or the QR code on the receipt. This allows SMRs to pick up the suspended meals without needing to show any other form of identification, which makes them more comfortable. Once the order is confirmed, the remaining number of meals for the day is updated on the bulletin board at the top of the tablet interface and the information on the kiosk is also synchronized.

7 SYSTEM EVALUATION

We conducted an evaluation of the FOODwell system concept and prototype, not only on the interactive kiosk interfaces but also on the interfaces used in restaurants. In addition to finding six potential users, we recruited six participants who have worked

in restaurants to participate in the evaluation of the FOODwell system prototype, as they play a crucial role in making the offer of suspended food sustainable and have rich experience in preparing suspended meals and interacting with SMRs. If the workflow is burdensome for them or potentially affects their business, they may not have the incentive to use the proposed design. During the evaluation, we asked the restaurant-experienced participants to perform various tasks using the prototype on a tablet, such as handling meal orders and completing transactions with SMRs. Both sets of participants were shown the kiosk interface used by SMRs, which was simulated on a tablet, given its similarity with the actual kiosk display. Throughout the process, we solicited their opinions and any concerns they had about the concept and the entire workflow.

We received much positive feedback that supports the concept and workflow, as well as the usability of FOODwell. In terms of usability, participants filled out the System Usability Scale (SUS) [8] after completing the tasks, and the score that FOODwell received was on average 90 (SD=8.164). Restaurant-experienced complimented the system's simple and intuitive interface on the restaurant tablet. More importantly, most participants consistently liked the concept of the system in how it helps SMRS through the clear and straightforward flow of reserving suspended meals through the kiosk. They were impressed by the integration of the identity verification mechanism of inserting the National Health Insurance card with the process of collecting suspended meals, which not only eases the psychological burden on recipients but also reduces the time spent waiting at the restaurant counter. Participants also mentioned that the reservation system can alleviate their concerns about missing out on suspended meals and offer greater flexibility in terms of pick-up time.

In addition to the positive feedback on the overall concept, we also received suggestions to improve usability, such as replacing some text with icons or changing the wording of certain menu items to match better with their mental model of the restaurant's existing workflow. We were also advised to consider handling normal meal orders through a switch or filter, in order to create a comprehensive meal ordering system.

8 CONCLUSION

To help a minority population struggling with hunger, some individuals are willing to provide free meals, known as suspended meals. However, existing suspended meal platforms have significant shortcomings, such as a lack of real-time updates for supply and demand, the difficulty for SMRs to access the platform due to lack of access to electronic devices, and neglect of SMRs' desire to keep their status as a minority confidential. These issues have made the suspended-meal system not fully leveraged as intended. To address these issues, we proposed FOODwell, a system that leverages existing kiosks in convenience stores and enables SMRs to reserve suspended meals through kiosks intuitively. It allows them to insert the NHI Card into the machine for qualification check and pick up reserved meals with a QR code receipt that makes the entire meal transaction look identical to those of other customers, preventing SMRs from revealing their minority status at the restaurant counter. Our system has been evaluated by participants resulting

in a SUS score of 90, which confirms its great usability and ease of use. We hope that FOODwell in the future will be a more trustworthy system for SMRs, providing them with a sense of certainty about meal availability, and feelings of comfort and relief when taking suspended meals, thereby allowing them to fully leverage the suspended-meal system and live without worrying about hunger.

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